

1.3 Propositional Equivalences

Friday, September 6, 2024 6:33 AM

Definition: A compound proposition which is always true is called a tautology. If it is always false, it is called a contradiction. Otherwise it is a contingency.

Consider $p \wedge \neg p$ and $p \vee \neg p$

Definition: Two compound propositions are logically equivalent if $p \leftrightarrow q$ is a tautology.

We denote this by $p \equiv q$.

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

How do we check that they are logically equivalent? Truth Tables!

Using De Morgan's Law, what is a logically equivalent way to say the negation of...

"I have a good nap or an afternoon coffee."

"I don't have a good nap and I don't have an afternoon coffee."
nor do I

Do you think that order matters when using conjunctions or disjunctions?

We may extend De Morgan's Laws

$$\neg(p_1 \wedge p_2 \wedge \dots \wedge p_n) \equiv \neg p_1 \vee \neg p_2 \vee \dots \vee \neg p_n$$

$$\neg(p_1 \vee p_2 \vee \dots \vee p_n) \equiv \neg p_1 \wedge \neg p_2 \wedge \dots \wedge \neg p_n$$

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$$\begin{aligned} \bigvee_{i=1}^n P_i &= P_1 \vee P_2 \vee \cdots \vee P_n \\ \bigwedge_{i=1}^n P_i &= P_1 \wedge P_2 \wedge \cdots \wedge P_n \end{aligned}$$

Think
summations

What should De Morgan's look like
in this notation?

[Even more practice! Show $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
This is proves a distributive property of $\wedge$ & $\vee$

Show $p \Rightarrow q \equiv \neg p \vee q$, then use this to write
 $(p \Rightarrow \neg q) \Rightarrow (\neg p \Rightarrow q)$ without implications

p	q	$p \Rightarrow q$	$\neg p \vee q$
T	T	T	T
F	T	T	T
T	F	F	F
F	F	T	T

$$\begin{aligned} (p \Rightarrow \neg q) \Rightarrow (\neg p \Rightarrow q) &\equiv (\neg p \vee \neg q) \Rightarrow (\neg p \vee q) \\ &\equiv (\neg p \vee \neg q) \Rightarrow (p \vee q) \equiv \neg(\neg p \vee \neg q) \vee (p \vee q) \\ &\equiv (p \wedge q) \vee (p \vee q) \end{aligned}$$

Tables 6, 7, 8 in the text (pg. 29)
are very useful for simplifying a
compound proposition and finding other
useful equivalent compound propositions.

Use a series of logical equivalences to
show $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$$

$$\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) \\
&\equiv \neg p \wedge (p \vee \neg q) \equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) \\
&\equiv \text{F} \vee (\neg p \wedge \neg q) \equiv \neg p \wedge \neg q
\end{aligned}$$